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The Rural Adult and Youth Skin protection study (RAYS): Protocol for a sun protection intervention trial in rural Utah and West Virginia sports programs

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ABSTRACT

Background: Children in rural and underserved communities face a higher risk of developing skin cancer later in life owing to social, behavioral, and environmental factors. The goal of the Rural Adult and Youth Sun Protection (RAYS) trial is to evaluate the efficacy of a multicomponent, multilevel sun-safety intervention in improving sun protection among children in these communities.

Methods: The RAYS cluster-randomized trial will test a sun safety intervention delivered through youth (ages 3–6) recreational sports settings in West Virginia and Utah and examine outcomes up to one year after enrollment. Enrolled sports leagues will be randomized to receive either RAYS, which targets sun safety-related adaptations at the individual, social, environmental, and policy levels, or an intervention that focuses on sports safety (no sun safety information). The primary outcome will be observed sun safety behaviors during sports practices and games; secondary outcomes will include parent-reported child sun protection and sunburn occurrence. Potential mediators (e.g., coach and parent knowledge and self-efficacy around skin cancer prevention) and factors related to trial outcomes, such as intervention fidelity and uptake and potential moderators (e.g., Area Deprivation Index), will also be examined.

Conclusions: The RAYS trial will examine the efficacy of a sports safety-focused intervention targeting sun protection behaviors among children in rural and underserved communities. The project will lead to new scientific understanding of theoretical, interventional, and implementation components that influence study outcomes and sociodemographic moderators, which will inform future intervention implementation in similar settings and identify communities which could best be served by RAYS.

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1. Background

Skin cancer affects more than 5 million Americans annually, and with increasing incidence potentially due to climate change [1–5]. Melanoma, the deadliest form of skin cancer, disproportionately affects rural areas: Incidence and mortality are 19% and 11% higher, respectively, among rural communities compared to urban communities [6–9], and this is reflected both nationally and in specific geographic areas (e.g., Utah). Skin cancer is often preventable if individuals limit ultraviolet radiation (UVR) exposure and avoid sunburn by using UVR protection strategies; this is particularly important in childhood when skin is more biologically sensitive to UVR damage, children spend substantial time outdoors including in organized sports, and early life sunburns are strongly associated with future skin cancer risk [10–14]. Relative to children in urban settings, children in rural and underserved (RAU) communities are at greater risk for skin cancer in adulthood: they have more frequent and intense outdoor exposures to UVR including during sports, suboptimal sun protection, and limited access to prevention resources [15–19]. Yet, no evidence-based sun safety interventions show durable effects on children's behavior in RAU areas, especially while playing outdoors. Sports leagues are a common community focal point, offering naturally-occurring opportunities for cancer education in settings where children have high UVR exposure [10,14,19–21]. Sports-based skin cancer prevention with young players can be impactful, as players are influenced by adult role models, need assistance with sun protection, and often lack cancer prevention education in other settings, such as school.

The goal of this trial is to examine the efficacy of a multicomponent, multilevel sun safety intervention (Rural Adult & Youth Sun [RAYS] protection program) with the intent of improving skin cancer prevention behaviors among young children living in RAU settings. RAYS is grounded in the Behavioral Ecological Model (BEM) [22], which integrates behavior change driven by environmental adaptation at the individual, social, and environmental levels. RAYS seeks to shape children's and coaches' sun safety behaviors by providing individual-level instruction and direct aide, social reinforcement of behaviors, and environmental supports. Participants in a pilot study of RAYS evaluated it as highly acceptable and the intervention demonstrated beneficial preliminary effects on observed child sun protective behaviors, coaches' sun protection behaviors, presence and use of sun safety supplies during practices and games, and an increase in knowledge and awareness of skin cancer prevention measures [23].

We will test the efficacy of RAYS in two rural U.S. states matched for high skin cancer burden and rurality Both Utah and West Virginia have high proportions of the state which are rural (77% and 78%, respectively) [15,24,25] and relatively high skin cancer incidence (41.4 per 100,000, 21.8 per 100,000, respectively; compared with the national average of 23.6 per 100,000) [16,17,26,27], including in rural areas [18,28,29]. The primary aim of this study is to evaluate the effectiveness of RAYS on the primary outcome of sun protection among children living in RAU communities. We will also examine multilevel parent, coach, team, and league mechanisms by which RAYS impacts youth sun protection. Finally, the trial will include an examination of the influence of intervention fidelity and uptake on RAYS' sun protection cognitions and child sun protection outcomes and examine dimensions of disadvantage as moderators in RAU communities.

2. Methods

2.1. Study design

This study is a 2-arm cluster-randomized trial taking place in Utah and West Virginia, in which developmental baseball leagues will be randomly assigned to RAYS vs. a control condition. Outcomes include directly-observed child sun safety behaviors after adjusting for environmental UVR before and after intervention, with parent-reported child sun protection and other endpoints collected through one year later. In addition, a process evaluation will be conducted, including exit interviews and fidelity checks, alongside the main outcome evaluation.

2.2. Participants and Setting

Local sports leagues will be eligible to participate if they convene baseball/softball teams or developmental baseball/softball programs (i.e., t-ball, coach pitch) for children ages 3–6 and are located in a rural county in Utah or West Virginia. Baseball and softball were selected due to youth sports being a central component of rural lifestyle and serving as a community focal point. Rurality will be defined using the Rural-Urban Commuting Area codes (RUCA) system [30]. Census tracts with a RUCA code $\geq$4 are considered rural [30]. Although skin cancers are more frequently diagnosed among individuals who identify as White and non-Hispanic, the states were matched demographically for other racial/ethnic groups as well as state population and geographic areas which will increase generalizability of findings.

2.3. Recruitment

We will recruit baseball/softball teams with children ages 3–6 (although it is possible children older than age 6 may participate) and parents/coaches/key informants ages 18+. Coaches will be eligible to participate if they are adults who serve as coaches or leaders of recreational and developmental sports serving children ages 3 and older. Parents will be eligible if they have a child age 3 or older participating in a sport, and they live and/or work in rural areas of Utah or West Virginia. Individuals will be eligible to participate in key informant interviews if they fit the criteria above as a parent or coach or are a league or community leader. Adults who do not speak or read English and individuals who have medical or other conditions (e.g., developmental delay) that would preclude their completion of surveys will be excluded.

The first level of recruitment will occur by league. Recreation departments and local baseball leagues will be identified in rural counties and contacted about participating. Study staff will share details about the study with league leadership and ensure the league includes divisions for children ages 3–6 years old and consist of four or more teams. We expect to enroll sports teams that include at least approximately 640 players across both arms (320 per arm).

Consent, enrollment, and randomization. Registration forms will be provided to administrators of leagues who agree to participate. These forms will gather basic information, such as season dates, if there are sun safety league policies, and whether coaches and parents are required to complete any safety training. Upon completion of registration, league administrators will be asked to provide contact information for RAYS-eligible coaches and player rosters. Leagues will be randomized to either the RAYS or control intervention using a matched randomization procedure [31] on a one-to-one basis accounting for within-state geographic proximity.

Coaches and their teams will be enrolled prior to the sports season during the player registration period. Coaches and players' parents will be sent the informed consent form via an electronic data collection platform (REDCap). Potential participants will review the form, provide informed consent, complete a registration form, and complete the baseline survey.

2.4. Assessment schedule

The baseline assessment will be sent to coaches and parents during the start of the sports season, usually in March through June depending on the league. Field observations and video recordings of coach and player behaviors completed by study staff will take place during the first two and last two games of the season. After the season has ended (6–8 weeks after start date), a post-intervention assessment will be sent to coaches and parents, as well as a second follow-up assessment one-year

following baseline. Surveys will be administered electronically by REDCap. The study team may also administer hard copies of the survey or items verbally if needed. Interviews with coaches and parents will be conducted after post-season follow-up assessment completion as part of a process evaluation.

2.5. Study procedures

After completing the baseline assessment and observation of at least one game or practice by study staff, coaches in both conditions will be invited to schedule a one-on-one intervention education session via videoconference. Training materials will include a digital presentation based on randomization – sun safety (intervention) or injury prevention (control). A copy of the presentation will be provided following the session in paper form or via email. Education sessions will be recorded for fidelity assessments. All coaches will be asked to share an educational video, similar to the one they viewed, with parents on their team along with a scripted message about team safety. Coaches will be trained and encouraged to model safety behaviors they learned and empower players to adopt either sun safety or injury prevention during games and practices. League leadership will be asked to share scripted safety messages throughout the season to support coaches' and parents' efforts learned in the educational materials. We provide a recommended schedule for sending safety messages weekly. However, we will encourage leagues to adapt for their community's needs. Coaches and parents will also receive outreach efforts and reminders to complete assessments (text, phone call, email) at the assessment times (baseline, post-intervention, 1-year post-baseline) [32].

2.6. Interventions

2.6.1. RAYS arm

League-level interventions. Leagues randomized to RAYS will receive a multilevel and multimodal intervention (Fig. 1). Before the season, they will be sent examples of sun safety policies their league could choose to adopt, such as scheduling games or practices outside of peak UVR hours or including sun safety recommendations on materials shared with parents. Leagues will be provided with environmental interventions, such as portable shade structures and banners prompting sun safety at ballfields. League leadership will also be provided with a media kit containing short messages about child sun protection which they will be encouraged to share through their existing channels, such as social media, newsletters, or existing email or text message threads to help strengthen community awareness and engagement. Leaders will be asked to reinforce sun safety's importance with coaches at meetings and through direct communication.

Coach-level interventions. Coaches in leagues assigned to RAYS will receive education on the relevance of skin cancer to children and in the sports setting, skin cancer prevention recommendations, behavioral and logistical strategies to promote children's use of sun protection, and ways to make sun safety a priority when outdoors. Educational content will include incidence and mortality rates of skin cancer in rural areas, how coaches can model sun protective behaviors for their team, and encourage coaches to instruct parents to pack their children's sun safety bag containing UVR protection supplies (Fig. 2). Coaches will receive instruction on how to explain the study-provided UVR wearables and what it demonstrates (i.e., a bracelet that changes color to indicate current sun exposure, and hence the need for use of sun protection) to parents and players, and will be encouraged to prompt players to use the UVR wearable at games and practices if allowed by the league. The UVR wearables do not display quantitative information on UVR dose and do

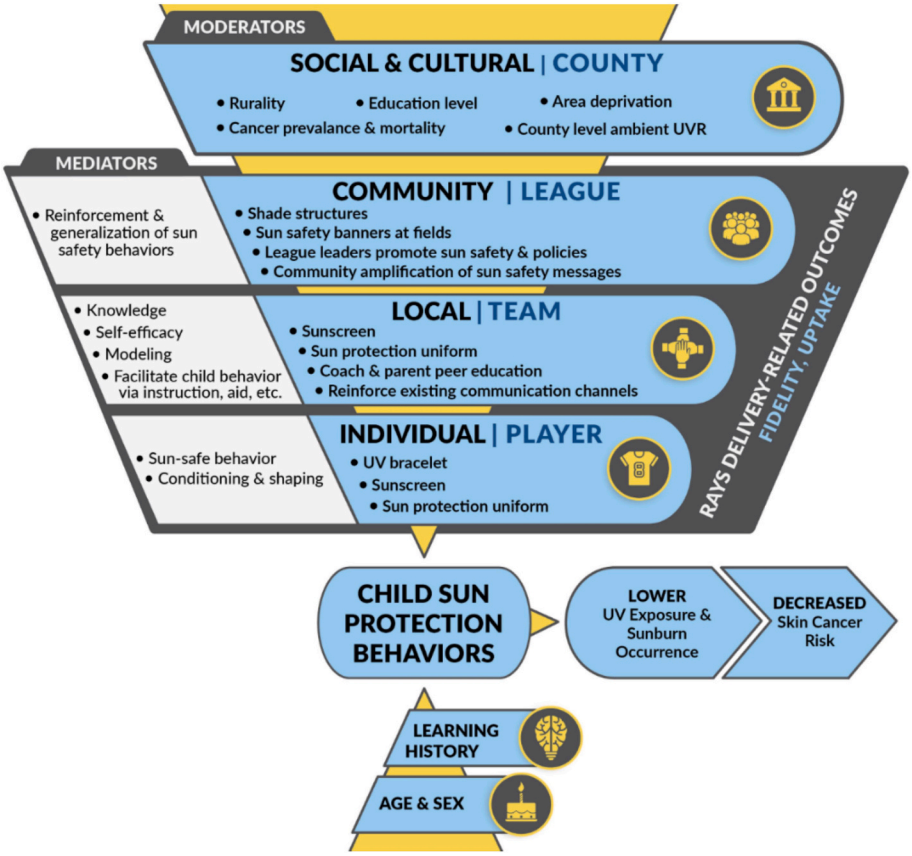


Fig. 1. RAYS conceptual framework.

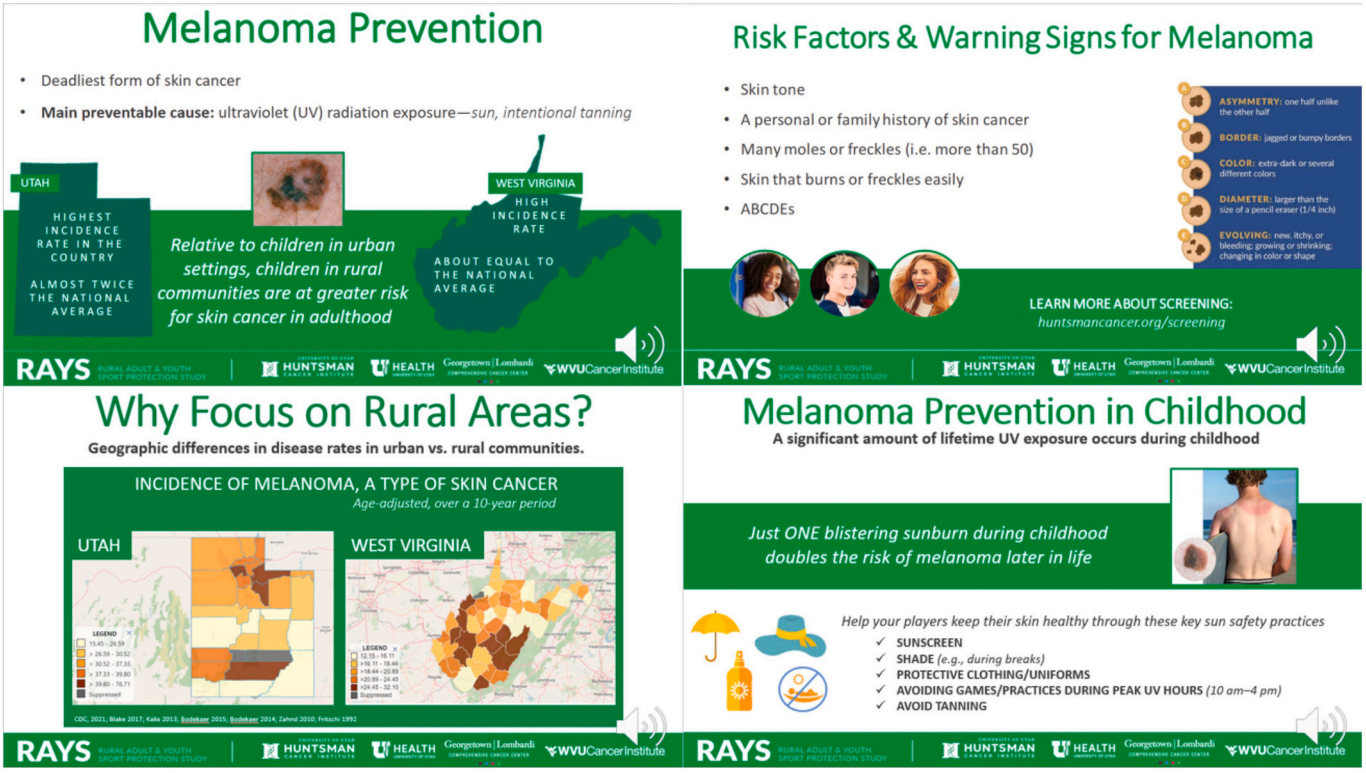


Fig. 2. Skin cancer prevention education content.

not yield downloadable data. To model safe behaviors for their team, coaches will receive sun safety supplies including sunglasses, UV protective sleeves, sunscreen, and a shade tent.

Parent-level intervention. Following the baseline assessment, parents will receive an email or text message from the coach that contains an educational presentation, similar to their training content, designed to encourage parents to model and support safe behaviors. They may also receive printed study information and education at games



Fig. 3. Injury prevention education content.

or practices. Each child on the team will receive a sun safety bag containing sunglasses, UV protective sleeves, UVR wearable, and sunscreen.

Control arm.

League-level interventions. Leagues randomized to the control arm will receive study banners for supplies and a media kit containing short messages about injury prevention for league leadership to share on league pages and social media sites for their coaches, players, and parents.

Coach-level interventions. Coaches will receive education on general injury prevention principles for child athletes based on publicly-available resources such as the Centers for Disease Control and Prevention and the American Academy of Pediatrics [33,34]. Content will include information on use and availability of protective gear (e.g., face masks, batting helmets), modeling use of protective equipment, teaching proper form (e.g., for batting, throwing, catching), providing warm-up and breaks during practices and games, and importance of cross-training through other physical activity to minimize overuse injuries (Fig. 3). Coaches will not receive any sun or sport safety materials other than education.

Parent-level interventions. Parents will receive an email or text message from the coach that contains an educational presentation on general injury prevention principles similar to the presentation viewed by their coaches.

2.7. Outcome Measures (Table 1)

Procedures were successfully tested in our pilot. The timepoints of assessment for all measures are summarized in Table 1.

Observed child sun protection (Primary Outcome). Study staff (one per game) will directly observe a select number of pre-intervention and post-intervention games (see Fig. 4). Staff will use a structured coding sheet to record observed sun-safe actions of coaches and players during game play. Coded behaviors include wearing a hat, sunglasses, UVR wearable, shirt with sleeves, long pants, any on-site sunscreen use (whether partial or full application), time spent in the shade, and coach encouragement of sun-safe behaviors. Furthermore, study staff will follow a protocol to record three 5-min wide-angle video segments (during warmup, when they play in-field, and play outfield). The videos will be uploaded to a secure server for subsequent coding for study-related outcomes. This coding will follow a similar process to the direct observation coding method described previously, which will include observers reviewing the same coded behaviors listed above. Concordance between direct and video observation will be assessed for quality control and to assure accurate measurement of outcomes.

Study staff will, for each player, tally the composite numbers of child sun protection behaviors exhibited at any time during the practice or game to form an index score. Other sun protection behaviors not included in the index score, such as use of the UVR wearable, will also be assessed.

Parent-reported child sun protection (Secondary Outcome). At baseline, post-intervention, and one-year post-baseline assessments (Fig. 4), all parents (regardless of trial arm) will report their child's use of sun protection (use and reapplication of sunscreen with an SPF of 30+; wearing shirt with sleeves, long pants, hat, or sunglasses; staying in the shade avoiding being in the sun between 10 am-4 pm; and spending time in the sun in order to get a tan or indoor tanning) over the last month using an adapted version of the Sun Habits Survey [32], a well-established and valid measure [35,36]. Parents will be asked to report on behavior during sports and at other times when outdoors [32]. Items will be modified to reflect the reporting period relevant for the current study (past month), to assess sun protection behaviors not included in the original Sun Habits Survey (e.g., sunscreen re-application).

Parent-report child sunburn (Secondary Outcome). At each assessment parents will report on the number of sunburns their children experienced (e.g., Since the beginning of the sports season, how many times did your child have a red or painful sunburn that lasted a day or more following a sports practice/game day?). Parents will also report their child's time (in hours) spent outdoors on weekends and weekdays during the summer.

Ambient UVR Levels (Covariate). Ambient UV levels will be measured when the study team attends observation games or practices. A handheld digital UV meter (Solarmeter Model 7.0 from Solar Light) will be used to measure UVA and UVB rays to provide an accurate UV reading for the precise area. These readings will be analyzed to determine exposure of teams. UV index data from a weather smartphone app (e.g., Weather Channel) will also be collected.

Table 1
Study measures & outcomes.

Measures	Timepoints for Assessment		
	Baseline	Post-intervention	One Year post-baseline
Primary outcome			
Observed child sun protection	✔	✔	
Secondary outcomes			
Parent-report child sun protection	✔	✔	✔
Parent-report child sunburn	✔	✔	✔
Objective ambient UVR	✔	✔	
Mediators			
UV bracelet frequency/duration of use	✔	✔	✔
Coach & parent knowledge	✔	✔	✔
Coach & parent self-efficacy	✔	✔	✔
Coach & parent facilitation	✔	✔	✔
Sun safety supplies & messaging	✔	✔	✔
League sun safety policy adoption	✔	✔	✔
Moderators			
Area-level: Area deprivation, skin cancer prevalence/mortality, ambient UVR	✔		
Individual-level: Rurality, education level	✔		

2.8. Putative mediators

2.8.1. Multilevel mediators of the intervention effects will be assessed at all assessments

UVR wearable frequency/duration of use. Duration (over what period of time) and frequency (how many times) of UVR wearable use at games/practices will be directly observed by study staff and reported by parents. This will serve as a manipulation check for adherence to the RAYS intervention and examined as a potential mediator for intervention effects. Parents will answer the question, “To the best of your knowledge, has your child ever worn a UVR color-changing sticker or bracelet that changes color when they are in the sun?”, on a “yes/no” scale.

Coach & parent knowledge. Parents and coaches will be asked to complete measures of their skin cancer prevention knowledge (i.e. “Melanoma is the deadliest form of skin cancer”, and “Exposure to sunlight is a major risk factor for melanoma”) by responding “true”, “false”, or “don't know”. These items were shown to be reliable in our RAYS pilot (alpha's > 0.85) and draw from the literature and our prior work [37–40].

Coach & parent self-efficacy. Coaches and parents will complete assessment questions about their self-efficacy for implementing and supporting sun protection strategies with their children/players (i.e. “use sun safety behaviors such as wearing sunscreen of SPF 30+ during sports practices/games, provide players with time to put on sunscreen during practices/games, encourage players to use sun safety behaviors”) [39]. Responses will be measured on a scale from 1 to 10, where 1 = Not confident at all and 10 = Extremely confident.

Coach & parent facilitation. Parents and coaches will be asked to report the frequency with which they engaged in behaviors to support child/player use of sun protection during baseball such as behavioral

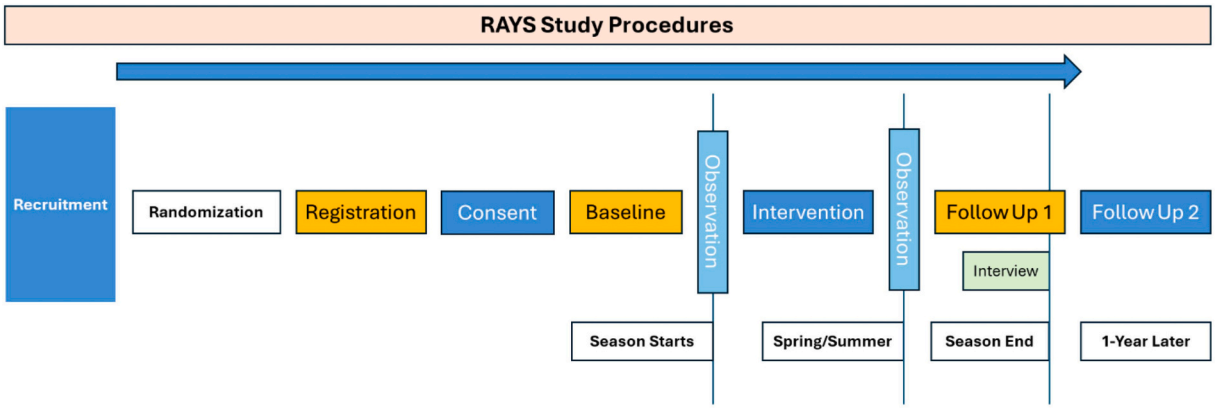


Fig. 4. RAYS study flow diagram.

modeling (their own use of sun protection strategies), facilitation of child sun protection via instruction (e.g., coach instructs all players to apply sunscreen before playing on the field), direct aide (e.g., parent helps child apply sunscreen), and behavioral reinforcement (e.g., coach provides praise to children who use sun protection) on a scale from 1 to 10, where 1 = Never and 10 = Every Game. Parent-reported sun protection has shown to be valid when compared to direct observation. Direct observation of these behaviors using the same procedures for the primary outcome of observed child sun protection behaviors will supplement these reports.

Sun safety supplies & messaging. Availability of sun safety supplies and recommended amplification of sun safety messages through league and community channels will be assessed by study staff using an environmental scan based on a modified standardized environmental measure for parks [41]. An environmental scan of each playing field will be conducted by study staff at all four observation games to assess the area's sun-protective conditions, regardless of trail arm.

League sun safety and policy adoption. At the beginning of the season and several months after the season has ended, league leadership will be asked to complete a survey that assesses if their league currently has any policies around sun safety ("Yes or "No"), if they share any information about sun safety and in what setting, if they *plan* to adopt any new sun safety policies, and how prepared their league feels to adopt these policies in future seasons ("Not at all ready" to "Extremely ready"). These assessment responses will be considered when study staff review each league's website and social media pages for evidence of sun safety messaging or adoption of sun safety policies influenced by the RAYS study. A subset of this data will be recorded and stored on REDCap to measure each league's change from pre-season to post-season.

2.9. Key informant interviews

At the end of each season, we will conduct key informant interviews to understand multi-level factors that may influence intervention effectiveness, community awareness of cancer prevention needs and priorities, and league and community goals. A subset of participating league staff, coaches, parents, and community members will be asked for feedback on their likes and dislikes about the study and how we could improve the acceptability of the program for future iterations of RAYS. We expect to recruit at least 20 coaches and 30 parents to complete interviews.

2.10. Moderators

We will explore moderators including rurality using the RUCA system [30], skin cancer prevalence and mortality using state cancer registry data [42,43], ambient UVR based on Environmental Protection Agency data sources [44], and household educational level reported by parents, along with other demographic characteristics. Potential area-level moderators of intervention effects will be investigated. The moderators will be measured on the block group or census tract level based on player ZIP Code. The Area Deprivation Index (ADI) [45,46] and its components will be examined as the primary moderator of interest because living in a disadvantaged neighborhood has been associated with poorer health outcomes [47–50]. The ADI, used as a composite measure of socioeconomic disadvantage, is a combination of 17 variables from the domains of education, income/employment, housing, and household characteristics.

2.11. Analytic approach

To evaluate the efficacy of RAYS, we will fit a linear mixed effects model of end-of-season sun protection behavior regressed on by RAYS intervention (coefficient of interest, Wald Test), baseline team composite index score (average across all players), ambient UVR at baseline and end-of-season, temperature, cloud cover, state, year, team (random effect), and league (random effect). If the model does not converge due to low team intracluster correlation (ICC), we will use randomization-based inference (RBI) to account for clustering while using a summary measure that adjusts for baseline covariates [51]. As secondary evaluations, we will compare individual sun protection behaviors, 3- and 12-month change in child behavior (parent-reported) from baseline, and sunburns [52,53].

To examine multilevel parent, coach, team, and league mechanisms by which RAYS impact youth sun protection, we will use Structural Equation Modeling (SEM)-based path analysis. Indirect effects will be estimated by product of coefficients under the causal potential outcomes framework. Path analysis models will be constructed according to the hypothesized framework with maximum likelihood parameter estimation [52,53]. We will calculate a 95% monte-carlo confidence interval of the product of coefficients and reject the null hypothesis if the interval excludes zero. Each model will adjust for the covariates without the random team or league effect. We will test mediating factors of parent, coach, and league-level mediators on team sun protection behaviors (alpha set at 0.05). To examine the influence of intervention fidelity and uptake on RAYS' sun protection cognitions and child sun protection outcomes, and to explore dimensions of disadvantage as moderators in underserved rural communities, we will include a RAYS by moderator interaction term for outcomes of interest (observed child sun protection behaviors, sunburns, end-of-season and 1 year change in parent-reported child sun protection behavior).

2.12. Power and sample size

Modern methods for handling missing data will be utilized (e.g., multiple imputation) to ensure statistical inferences are valid under

missingness-at-random (data values and missingness pattern are independent given the observed data), the most general data-centric missing data assumption. In the most conservative setting, we will enroll $N = 64$ teams with $N = 10$ players each and assume an intra-class correlation (ICC) of 0.03, and within-team outcome correlation of 0.50. With at least $N = 640$ players, the difference in mean player index score is assumed to be normally distributed. At alpha of 0.05, we are 80% powered to detect an effect size of 0.56 (Cohen's d, moderate). More realistically, we expect a smaller ICC at the league-level (ICC of 0.01) and more moderate within-team correlation among outcomes (0.30). In this scenario we are 80% powered to detect an effect size of 0.47. These effect sizes are consistent with sun protection behaviors observed in the pilot, and we expect strong effect sizes in the composite score as seen in prior studies [54]. In the plausible setting of ICC of 0.005 and within-team correlation of 0.20, we are 80% powered to detect an effect size of 0.38.

We will enroll at least $N = 200$ parents and anticipate at least 83% completion rate. For parents, if the total, standardized RAYS effect is 0.26, we are 80% powered to detect an indirect effect of 0.17. For coaches, if the total standardized RAYS effect is 0.33, we are 80% powered to detect an indirect effect of 0.22. Power will further increase by adjusting for planned covariates. As moderators, we do not expect to detect statistical significance for the effect sizes planned in the primary analysis; however, these are key metrics to inform broader dissemination. Under conservative settings (ICC of 0.03 and within-team correlation of 0.50) the following effect sizes can be detected with 80% power: 0.83 (league−/community-level moderator), 1.07 (coach-level moderator), and 1.51 (parent-level moderator).

3. Discussion

RAYS will be one of the first studies to test a skin cancer prevention program for youths delivered through a community setting focused on rural populations who experience disparities in skin cancer incidence and may have inadequate resources to address skin cancer prevention [55–57]. As highlighted in a recent portfolio analysis of National Cancer Institute-supported research [58], efforts to address skin cancer in rural areas are greatly needed. RAYS helps address this need by targeting young children in a novel prevention setting in which they are at risk for high UVR exposure and which has not been adequately studied. In addition, mediator analyses will examine mechanisms of action for RAYS' effects, which will guide further intervention development (e.g., augmented intervention components targeting mechanisms of change). Moderator analyses will determine subgroups requiring tailored versions of RAYS that are more modifiable to better meet communities' needs.

Strengths of this study include its design as a cluster-randomized trial across multiple rural counties in two different U.S. states, longitudinal design with multiple assessments, attention to seasonal and environmental effects, and objective measurement of UVR exposure. In addition, the trial embraces constructs such as fidelity, consistent with the Reach, Effectiveness, Adoption, Implementation, and Maintenance (RE-AIM) Framework [59] to inform real-world scale-up and outside the context of a trial. Limitations of the study include that it may not be generalizable outside of certain geographic areas, and although the study targets rural areas, there may be limited demographic variability in the samples recruited. Despite these limitations, this trial focuses on individuals who face high skin cancer risk, those from rural and underserved areas.

If RAYS is found to be efficacious in improving child sun protection, we will seek to evaluate the implementation of RAYS in additional communities under real-world conditions to promote the systematic uptake of the trial's findings and other evidence-based practices on youth sun safety into developmental baseball programs. It may also be possible to expand into other youth sports with high UVR exposure, as lessons learned from this trial will guide and inform youth skin cancer prevention and facilitate translation.

CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

[1] G.P. Guy Jr., S.R. Machlin, D.U. Ekwueme, K.R. Yabroff, Prevalence and costs of skin cancer treatment in the US, 2002–2006 and 2007–2011, Am. J. Prev. Med. 48 (2) (2015) 183–187, https://doi.org/10.1016/j.amepre.2014.08.036.

[2] H.W. Rogers, M. Weinstock, A.R. Harris, et al., Incidence estimate of nonmelanoma skin cancer in the United States, 2006, Arch. Dermatol. 146 (3) (2010) 283–287, https://doi.org/10.1001/archdermatol.2010.19.

[3] C. Fitzmaurice, D. Abate, N. Abbasi, et al., Global, regional, and national cancer incidence, mortality, years of life lost, years lived with disability, and disability-adjusted life-years for 29 cancer groups, 1990 to 2017: a systematic analysis for the global burden of disease study, JAMA Oncol. 5 (12) (2019) 1749–1768, https://doi.org/10.1200/jco.2018.36.15_suppl.1568.

[4] B. Diffey, Climate change, ozone depletion and the impact on ultraviolet exposure of human skin, Phys. Med. Biol. 49 (1) (2004) R1–11.

[5] A.K. Bharath, R.J. Turner, Impact of climate change on skin cancer, J. R. Soc. Med. 102 (6) (2009) 215–218, https://doi.org/10.1258/jrsm.2009.080261.

[6] K.D. Blake, J.L. Moss, A. Gaysynsky, S. Srinivasan, R.T. Croyle, Making the case for investment in rural cancer and control: an analysis of rural cancer incidence, mortality and funding trends, Cancer Epidemiol. Biomarkers Prev. 26 (7) (2017) 992–997, https://doi.org/10.1158/1055-9965.EPI-17-0092.

[7] T.L. Bowles, C. Sweeney, J. Snyder, et al., Impact of rurality on melanoma diagnosis in Utah, Melanoma Manag. 8 (2) (2021) MMT56, https://doi.org/10.2217/mmt-2020-0023.

[8] W.E. Zahnd, A.J. Fogleman, W.D. Jenkins, Rural-urban disparities in stage of diagnosis among cancers with preventive opportunities, Am. J. Prev. Med. 54 (5) (2018) 688–698, https://doi.org/10.1016/j.amepre.2018.01.021.

[9] Surveillance Epidemiology and End Results Program, SEER Stat Fact Sheets: Melanoma of the Skin, National Cancer Institute, 2026. Accessed August 27, 2022, https://seer.cancer.gov/statfacts/html/melan.html.

[10] E. Mahé, A. Beauchet, M. de Paula Corrêa, et al., Outdoor sports and risk of ultraviolet radiation-related skin lesions in children: evaluation of risks and prevention, Br. J. Dermatol. 165 (2) (2011) 360–367, https://doi.org/10.1111/j.1365-2133.2011.10415.x.

[11] S.J. Balk, Ultraviolet radiation: a hazard to children and adolescents, Pediatrics 127 (3) (2011) e791–e817, https://doi.org/10.1542/peds.2010-3502.

[12] A.C. Green, S.C. Wallingford, P. McBride, Childhood exposure to ultraviolet radiation and harmful skin effects: epidemiological evidence, Prog. Biophys. Mol. Biol. 107 (3) (2011) 349–355, https://doi.org/10.1016/j.pbiomolbio.2011.08.010.

[13] L.R. Larson, G.T. Green, H.K. Cordell, Children's time outdoors: results and implications of the National Kids Survey, J. Park. Recreat. Adm. 29 (2) (2011) 1–20.

[14] Aspen Institute, Project Play: State of Play, Accessed November 8, 2021, https://www.aspenprojectplay.org/state-of-play-2021, 2021.

[15] S. O'Leary, R. Wilson, T. Boettner, The State of Rural West Virginia. https://wvpolicy.org/wp-content/uploads/2018/10/WVCBP_RuralWV_2018.pdf, 2018.

[16] P. Aggarwal, P. Knabel, A.B. Fleischer Jr., United States burden of melanoma and non-melanoma skin cancer from 1990 to 2019, J. Am. Acad. Dermatol. 85 (2) (2021) 388–395, https://doi.org/10.1016/j.jaad.2021.03.109.

[17] Centers for Disease Control and Prevention, Rate of New Cancers: Melanoma of the Skin, Accessed July 19, 2022, https://gis.cdc.gov/cancer/USCS/DataViz.html, 2026.

[18] WV Cancer Registry, WV Department of Health and Human Resources, West Virginia University Cancer Institute, West Virginia Cancer Burden Report, February 2020, https://oeps.wv.gov/cancer/documents/data/burdenreport2019.pdf, 2019.

[19] Y.P. Wu, B. Parsons, Y. Jo, et al., Outdoor activities and sunburn among urban and rural families in a Western region of the US: implications for skin cancer prevention, Prev. Med. Rep. 29 (2022) 101914, https://doi.org/10.1016/j.pmedr.2022.101914.

[20] K.A. Ferris, B. Oosterhoff, A. Metzger, Organized activity involvement among rural youth: gender differences in associations between activity type and developmental outcomes, J. Res. Rural. Educ. 28 (15) (2013) 1–16.

[21] B.J. Bishop, S.J. Coakes, P.N. D'Rozario, Sense of community in rural communities, in: A.T. Fisher, C.C. Sonn, B.J. Bishop (Eds.), Psychological Sense of Community: Research, Applications, and Implications, Springer US, 2002, pp. 271–290.

[22] M. Hovell, D. Wahlgren, M. Adams, The logical and empirical basis for the behavioral ecological model, in: R.J. Diclemente, R.A. Crosby, M.C. Kegler (Eds.), Emerging Theories in Health Promotion Practice and Research, Jossey-Bass/Wiley, 2009, pp. 415–449, chap 15.

[23] Y.P. Wu, H.L. Brady, A.P. Wankier, et al., Preliminary effects of a rural skin cancer prevention intervention for youths, Health Psychol. 43 (5) (2024) 339–351.

[24] Kem C. Gardner Policy Institute, Regional Fact Sheet, Utah Legislative Policy Summit, 2007. https://gardner.utah.edu/wp-content/uploads/RuralUtahSheet.pdf?x71849.

[25] West Virginia Department of Health and Human Resources, Creating a Culture of Health in Rural West Virginia: State Rural Health Plan 2018-2022. https://dhhr.wv.gov/ruralhealth/publicationsandresources/Documents/2018%20State%20Rural%20Health%20Plan%20(Final).pdf, 2018.

[26] A.C. Society, Melanoma of the Skin Cancer Statistics, Accessed May 5, 2026, https://cancerstatisticscenter.cancer.org/types/melanoma-of-the-skin, 2026.

[27] M.M. Millar, K. Herget, C. Howlett, R. Codden, J.A. Doherty, Cancer in Utah: Incidence and Mortality Statistics through 2021, 2024.

[28] Health UDo, Public Health Indicator Based Information System (IBIS), Updated 2/2016. Accessed 3/3/2016, 2016, https://ibis.health.utah.gov/indicator/Introduction.html, 2026.

[29] Utah Comprehensive Cancer Control Program, 2021–2025 Utah Comprehensive Cancer Prevention and Control Plan, 2021 file:///C:/Users/u0474850/Downloads/2021%20state%20cancer%20plan%20finalFINAL_for508.pdf.

[30] WWAMI Rural Health Research Center, Rural Urban Commuting Area Codes, WWAMI Rural Health Research Center, 2026, p. 1017. http://depts.washington.edu/uwruca/.

[31] R. Greevy, B. Lu, J.H. Silber, P. Rosenbaum, Optimal multivariate matching before randomization, Biostatistics 5 (2) (2004) 263–275, https://doi.org/10.1093/biostatistics/5.2.263.

[32] K. Glanz, A.L. Yaroch, M. Dancel, et al., Measures of sun exposure and sun protection practices for behavioral and epidemiologic research, Arch. Dermatol. 144 (2) (2008) 217–222, https://doi.org/10.1001/archdermatol.2007.46.

[33] Protect the Ones You Love: Sports Injuries. https://www.cdc.gov/safechild/fact_sheets/sports-injury-fact-sheet-english-a.pdf, 2023.

[34] American Academy of Pediatrics, Healthy Children: Sports Injury Prevention Tips, Accessed July, 2022, https://www.healthychildren.org/English/health-issues/injuries-emergencies/sports-injuries/Pages/Sports-Injuries-Treatment.aspx, 2026.

[35] D.L. O'Riordan, E. Nehl, P. Gies, et al., Validity of covering-up sun-protection habits: association of observations and self-report, J. Am. Acad. Dermatol. 60 (5) (2009) 739–744, https://doi.org/10.1016/j.jaad.2008.12.015.

[36] K. Glanz, F. McCarty, E.J. Nehl, et al., Validity of self-reported sunscreen use by parents, children, and lifeguards, Am. J. Prev. Med. 36 (1) (2009) 63–69.

[37] R.E. Glasgow, S.M. Harden, B. Gaglio, et al., RE-AIM planning and evaluation framework: adapting to new science and practice with a 20-year review, Front. Public Health 7 (2019) 64, https://doi.org/10.3389/fpubh.2019.00064.

[38] E. Proctor, H. Silmere, R. Raghavan, et al., Outcomes for implementation research: conceptual distinctions, measurement challenges, and research agenda, Admin. Pol. Ment. Health 38 (2) (2011) 65–76, https://doi.org/10.1007/s10488-010-0319-7.

[39] M.K. Tripp, P.M. Diamond, S.W. Vernon, P.R. Swank, P. Dolan Mullen, E.R. Gritz, Measures of parents' self-efficacy and perceived barriers to children's sun protection: construct validity and reliability in melanoma survivors, Health Educ. Res. 28 (5) (2013) 828–842, https://doi.org/10.1093/her/cys114.

[40] Y.P. Wu, E. Nagelhout, L.G. Aspinwall, et al., A novel educational intervention targeting melanoma risk and prevention knowledge among children with a familial risk for melanoma, Patient Educ. Couns. 101 (3) (2017) 452–459, https://doi.org/10.1016/j.pec.2017.10.008.

[41] Community Park Audit Tool (CPAT) Version 3, 2012.

[42] National Cancer Institute, SEER Incidence Data, 1975–2019, Accessed August 8, 2022, https://seer.cancer.gov/data/, 2026.

[43] National Cancer Institute, NCI Cancer Atlas, Accessed August 8, 2022, https://gis.cancer.gov/canceratlas/, 2026.

[44] United States Environmental Protection Agency: Sun safety, United States Environmental Protection Agency, Accessed June 21, 2018, https://www.epa.gov/sunsafety, 2026.

[45] A.J.H. Kind, W.R. Buckingham, Making Neighborhood-disadvantage metrics accessible — the Neighborhood atlas, N. Engl. J. Med. 378 (26) (2018) 2456–2458, https://doi.org/10.1056/NEJMp1802313.

[46] Center for Health Disparities Research at the University of Wisconsin, About Neighborhood Atlas, 2026. https://www.neighborhoodatlas.medicine.wisc.edu.

[47] B.G. Link, J. Phelan, Social conditions as fundamental causes of disease, J. Health Soc. Behav. Spec No (1995) 80–94.

[48] A.J. Kind, S. Jencks, J. Brock, et al., Neighborhood socioeconomic disadvantage and 30-day rehospitalization: a retrospective cohort study, Ann. Intern. Med. 161 (11) (2014) 765–774, https://doi.org/10.7326/m13-2946.

[49] P.M. Lantos, K. Hoffman, S.R. Permar, et al., Neighborhood disadvantage is associated with high cytomegalovirus seroprevalence in pregnancy, J. Racial Ethn. Health Disparities 5 (4) (2018) 782–786, https://doi.org/10.1007/s40615-017-0423-4.

[50] J. Hu, A.J.H. Kind, D. Nerenz, Area deprivation index predicts readmission risk at an urban teaching hospital, Am. J. Med. Qual. 33 (5) (2018) 493–501, https://doi.org/10.1177/1062860617753063.

[51] W.F. Rosenberger, D. Uschner, Y. Wang, Randomization: the forgotten component of the randomized clinical trial, Stat. Med. 38 (1) (2019) 1–12, https://doi.org/10.1002/sim.7901.

[52] J. Pearl, Causal inference in statistics: an overview, Stat. Surv. 3 (none) (2009) 96–146, 51, https://doi.org/10.1214/09-ss057.

[53] D. Tofighi, D.P. MacKinnon, Monte Carlo confidence intervals for complex functions of indirect effects, Struct. Equ. Model. 2 (23) (2016) 194–205, https://doi.org/10.1080/10705511.2015.1057284.

[54] A.J. Dietrich, A.L. Olson, C.H. Sox, T.D. Tosteson, J. Grant-Petersson, Persistent increase in children's sun protection in a randomized controlled community trial, Prev. Med. 31 (5) (2000) 569–574, https://doi.org/10.1006/pmed.2000.0746.

[55] C.D. Phillips, K.R. McLeroy, Health in rural America: remembering the importance of place, Am. J. Public Health 94 (10) (2004) 1661–1663, https://doi.org/10.2105/ajph.94.10.1661.

[56] A.H. Hansell, P.R. Giacobbi Jr., D.K. Voelker, A scoping review of sport-based health promotion interventions with youth in Africa, Health Promot. Pract. 22 (1) (2021) 31–40, https://doi.org/10.1177/1524839920914916.

[57] D.L. Elliot, L. Goldberg, E.L. Moe, C.A. DeFrancesco, M.B. Durham, H. Hix-Small, Preventing substance use and disordered eating: initial outcomes of the ATHENA (athletes targeting healthy exercise and nutrition alternatives) program, Arch.

Pediatr. Adolesc. Med. 158 (11) (2004) 1043–1049, https://doi.org/10.1001/archpedi.158.11.1043.
[58] A. Gaysynsky, S.J. Weaver, R.C. Vanderpool, et al., Advancing rural cancer control: a portfolio analysis of research funded by the National Cancer Institute, 2016–2024, J. Rural. Health 42 (1) (2026) e70120, https://doi.org/10.1111/jrh.70120.
[59] R.E. Glasgow, T.M. Vogt, S.M. Boles, Evaluating the public health impact of health promotion interventions: the RE-AIM framework, Am. J. Public Health 89 (9) (1999) 1322–1327, https://doi.org/10.2105/AJPH.89.9.1322.